

Data import & Data organization in spreadsheets

ds4owd - data science for openwashdata

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 ds4owd-002.github.io/website/

Learning Objectives (for this week)

1. Learners can import data from files in CSV and XLSX format located in sub-directories of the root directory.
2. Learners can explain the difference between the vector class character and the vector class factor.
3. Learners can discuss the difference between unprocessed raw data, processed analysis-ready data, and data underlying a publication.
4. Learners can apply 12 principles for data organisation in spreadsheets to the layout of a provided dataset.

Homework module 3

Task 3: Data for a country of your choice

- for the country you live or work in
- for the year 2000 and 2020
- for all variables that are not “safely managed sanitation services”

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2   filter(iso3 == "UGA",
3         year %in% c(2000, 2020),
4         varname_short != "san_sm")
```

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```

```
1 sanitation_uga |>
2   count(iso3, year, varname_short)
```

iso3	year	varname_short	n
UGA	2000	san_bas	3
UGA	2000	san_lim	3
UGA	2000	san_od	3
UGA	2000	san_unimp	3
UGA	2020	san_bas	3
UGA	2020	san_lim	3
UGA	2020	san_od	3
UGA	2020	san_unimp	3

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```

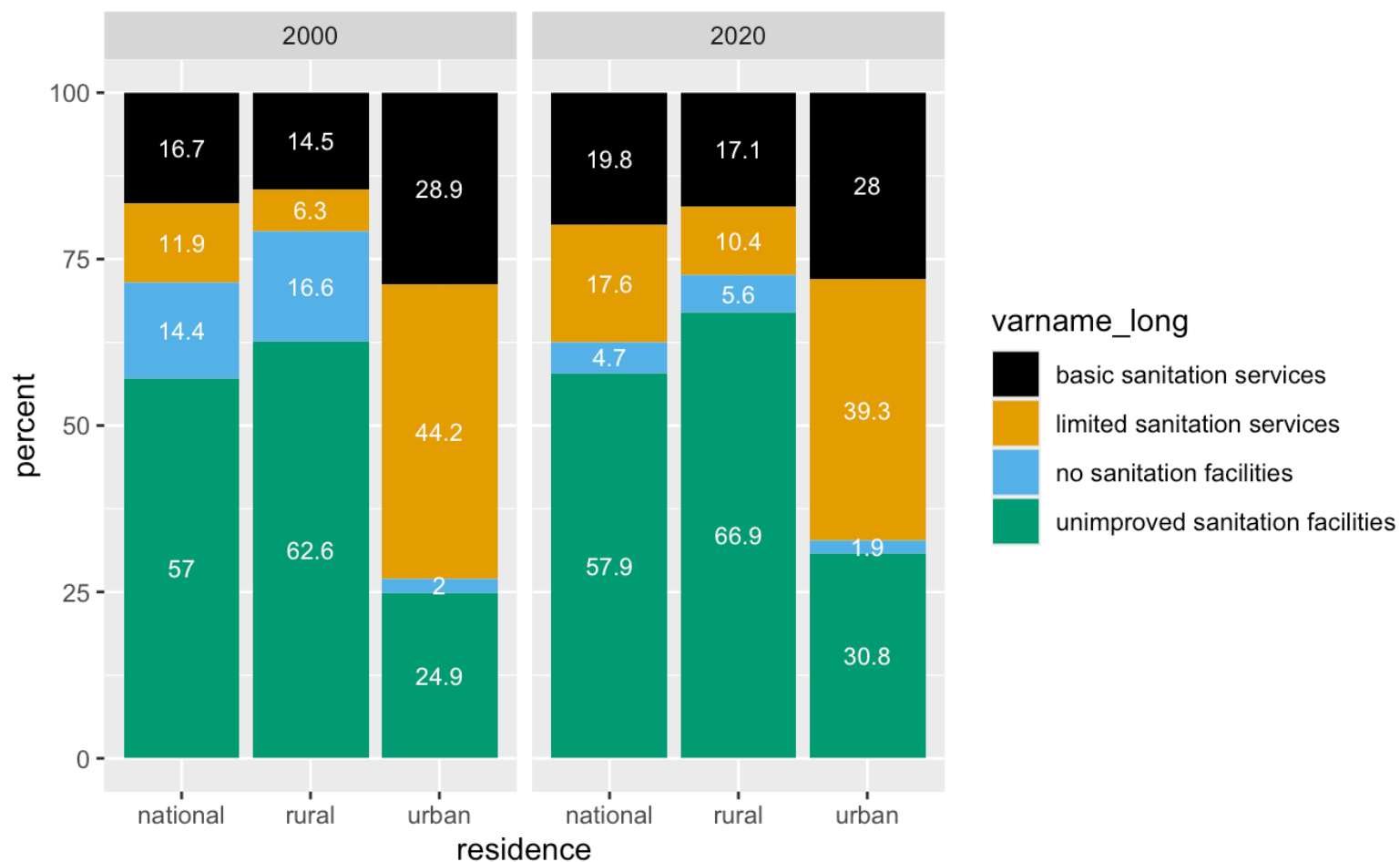
iso3	year	varname_short	n
------	------	---------------	---

```
# A tibble: 0 × 4
# i 4 variables: iso3 <chr>, year <dbl>,
# varname_short <chr>, n <int>
```

- One row cannot have two values (2000 and 2020) for the same variable
- One year cannot 2000 & 2020 at the same time
- One year is either 2000 or 2020

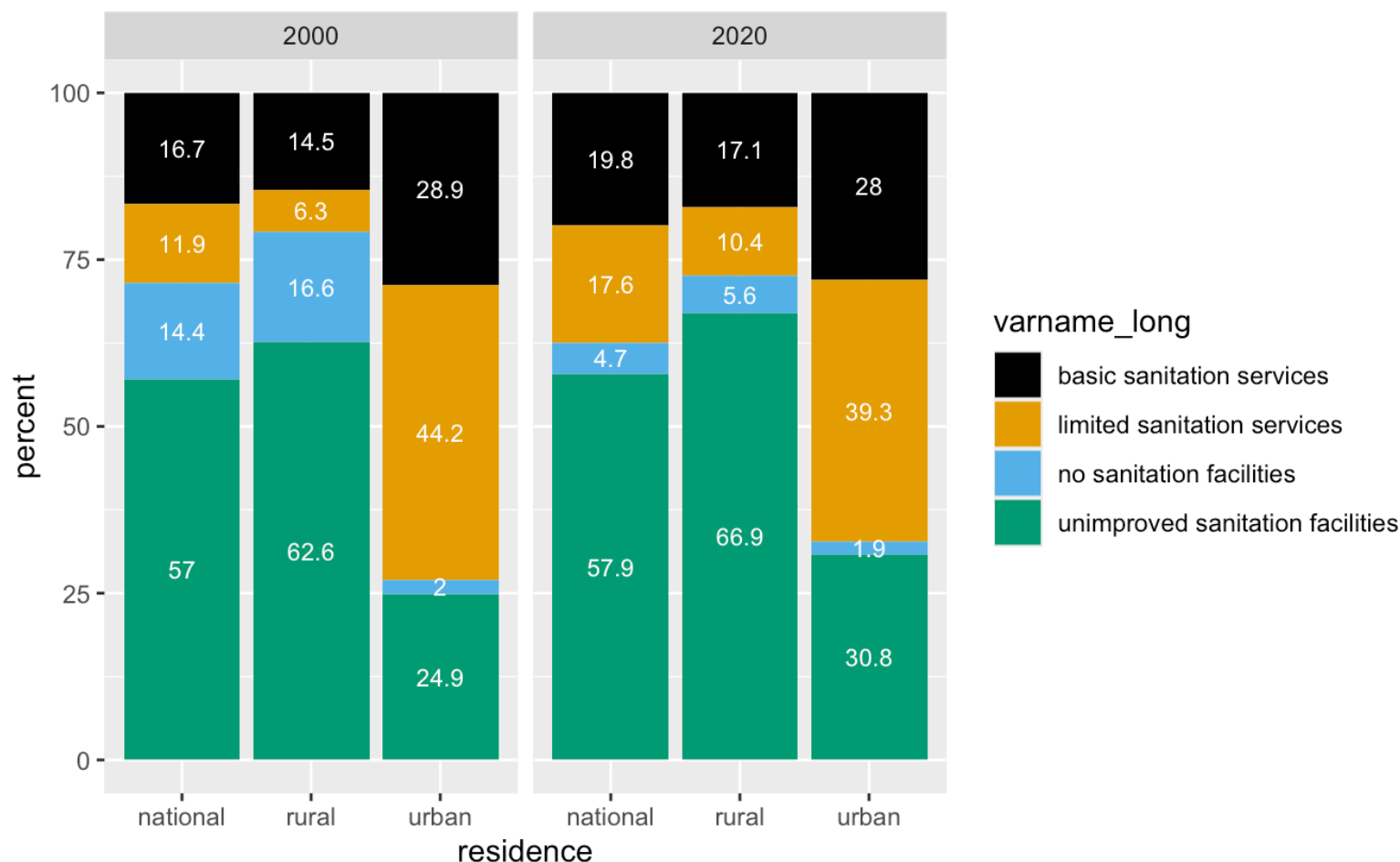
Task 5 & 6: Make a plot & inspect it

1. Look at the plot that you created. What do you notice about the order of the bars / order of the legend?
2. What would you want to change?
3. Why did we remove “safely managed sanitation services” from the data set in Task 3?



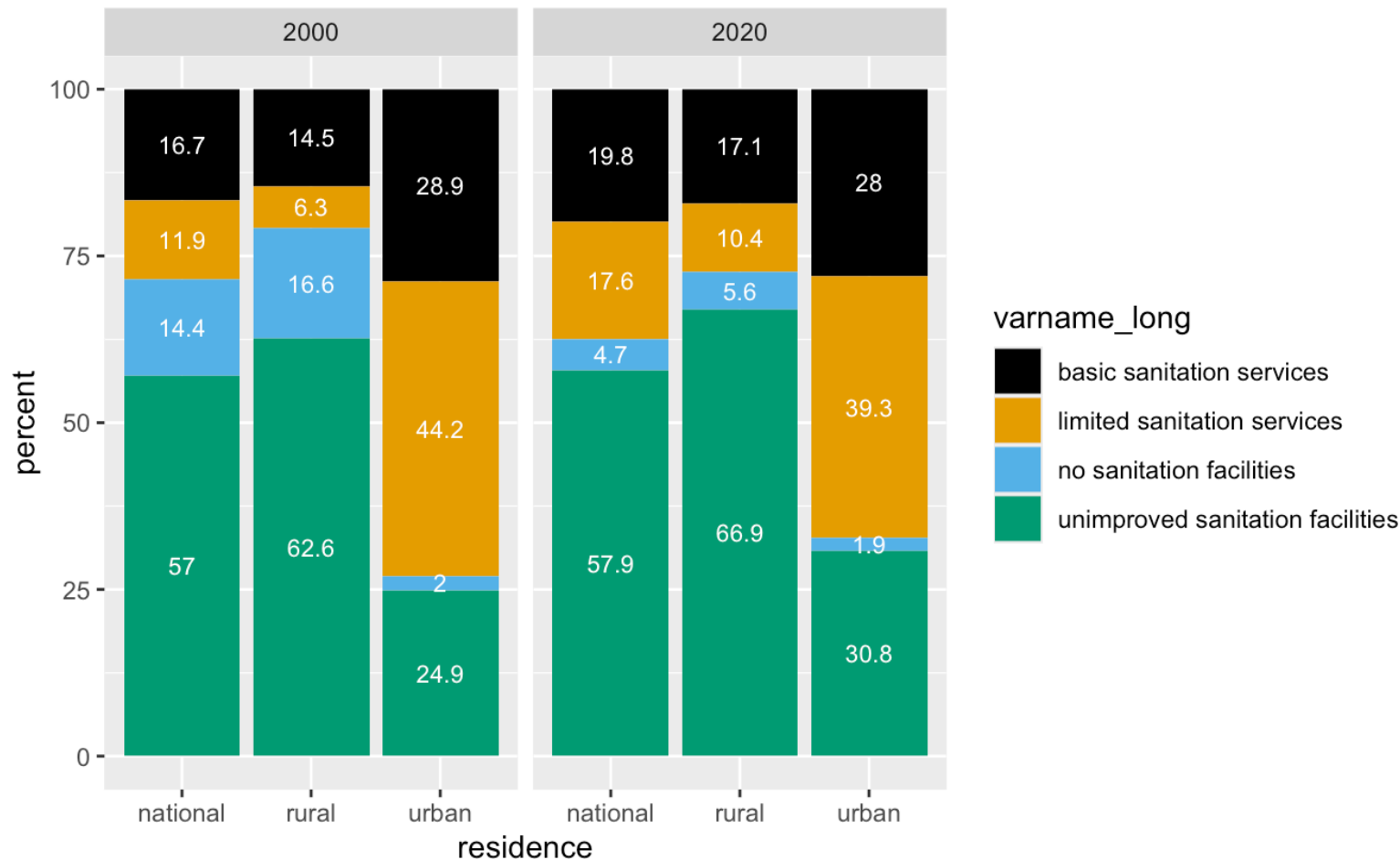
Task 5 & 6: Make a plot & inspect it

1. Look at the plot that you created. What do you notice about the order of the bars / order of the legend? **alphabetical order**
2. What would you want to change? put in order of the “sanitation ladder”
3. Why did we remove “safely managed sanitation services” from the data set in Task 3?



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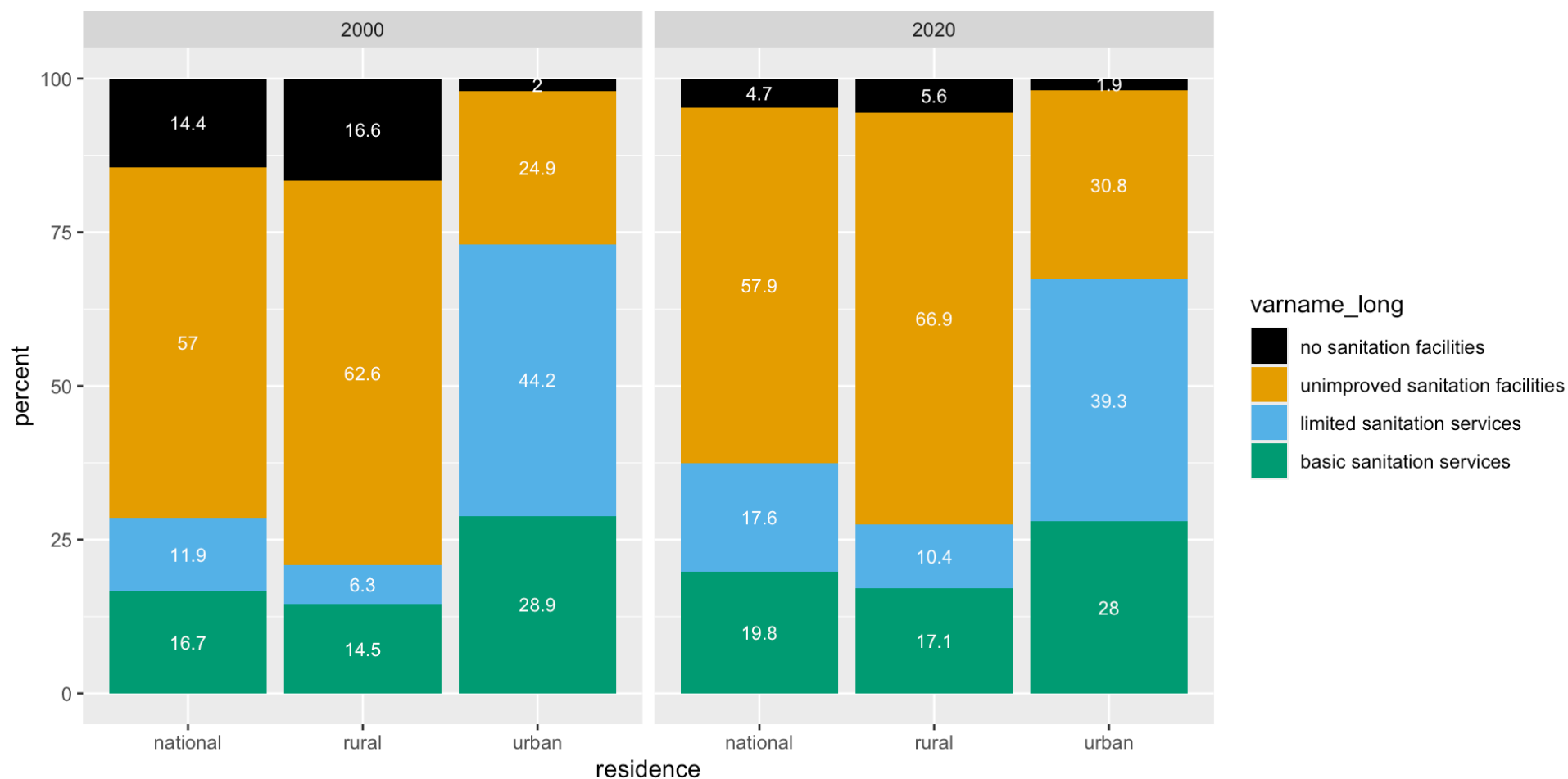


Sanitation ladder?

varname_short	varname_long	simplified
san_sm	safely managed sanitation services	a decent toilet that's not shared, and where pee & poo safely moved & treated
san_bas	basic sanitation services (improved sanitation facilities which are not shared)	a decent toilet that's not shared
san_lim	limited sanitation services (improved sanitation facilities which are shared)	a decent toilet that's shared
san_unimp	unimproved sanitation facilities	an inadequate toilet
san_od	no sanitation facilities (open defecation)	no toilet

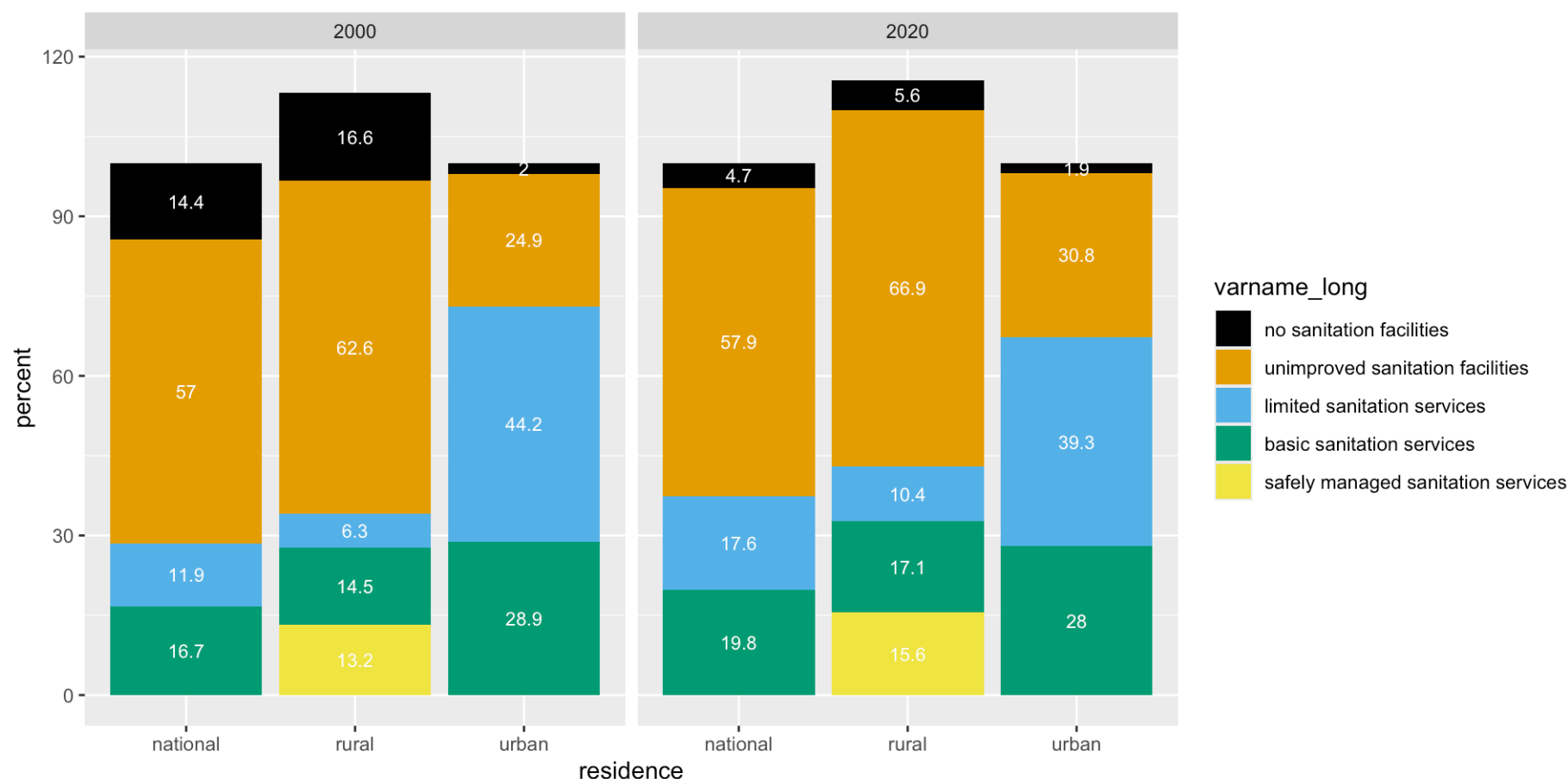
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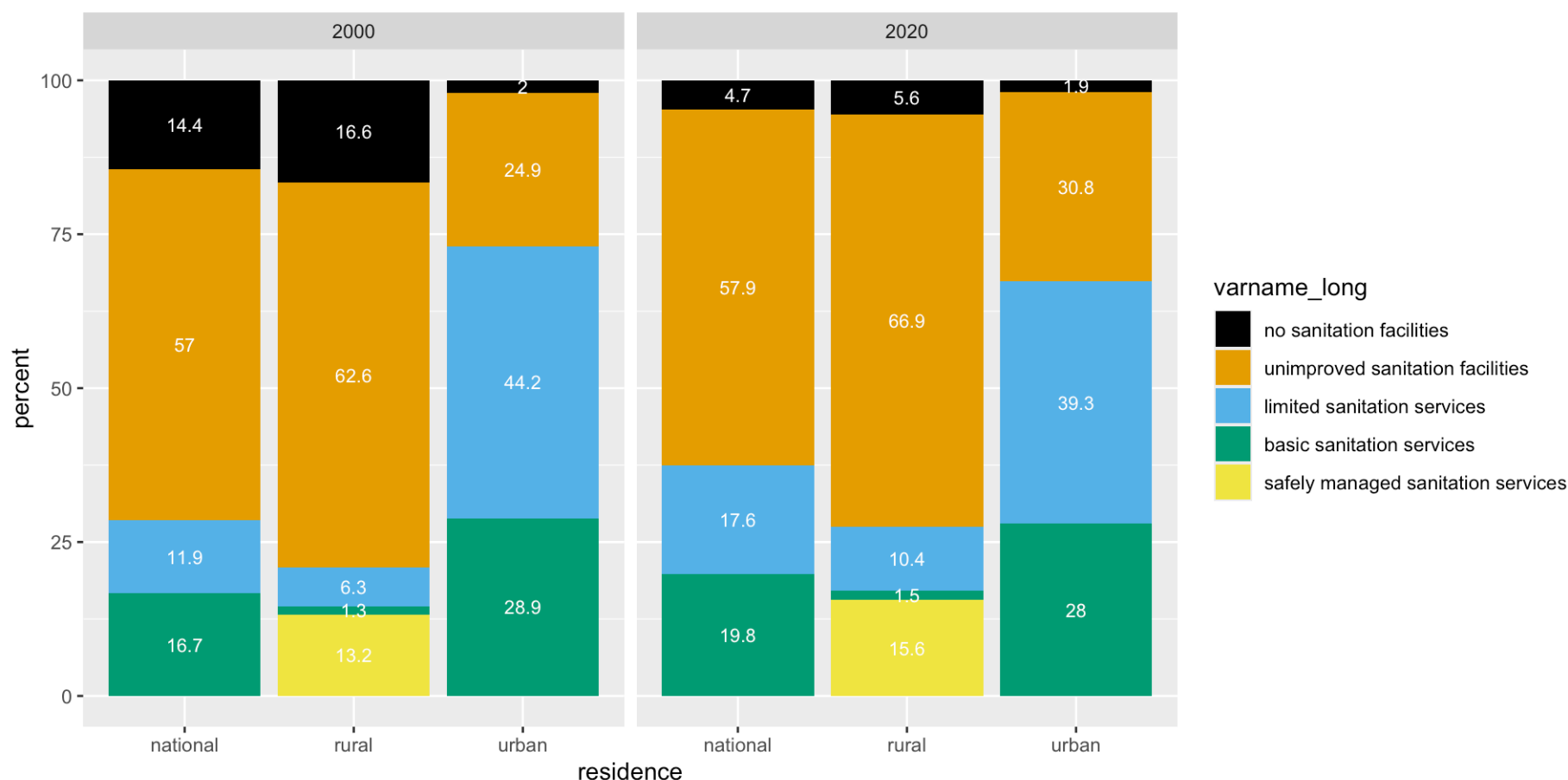
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Types of variables - Remember?

numerical

discrete variables

- non-negative
- whole numbers
- e.g. number of students, roll of a dice

continuous variables

- infinite number of values
- also dates and times
- e.g. length, weight, size

non-numerical

categorical variables

- finite number of values
- distinct groups (e.g. EU countries, continents)
- **ordinal** if levels have natural ordering (e.g. week days, school grades)

Factors in R

My turn: Factors in R

Sit back and enjoy!

Take a break

Please get up and move! Let your emails rest in peace.



Your turn: md-04-exercises

1. Open posit.cloud/spaces/663318/content in your browser.
2. Verify that the ds4owd workspace is open.
3. Click Start next to md-04-exercises.
4. In the File Manager in the bottom right window, locate the 02a-factors-your-turn.qmd file and click on it to open it in the top left window.

Data import

Reading rectangular data into R



CSV & XLSX

readr

- `read_csv()` - comma delimited files
- `read_csv2()` - semicolon separated files
(common in countries where , is used as the decimal place)
- `read_tsv()` - tab delimited files
- `read_delim()` - reads in files with any delimiter
- ...

readxl

- `read_excel()` - read xls or xlsx files
- ...

Reading data from CSV files

- import unprocessed raw data

```
1 waste <- read_csv("data/raw/waste-city-level.csv")
2
3 waste
```

A tibble: 367 × 113

	iso3c	region_id	country_name	income_id	city_name	additional_data_annu... ¹
	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
1	AFG	SAS	Afghanistan	LIC	Jalalabad	<NA>
2	AFG	SAS	Afghanistan	LIC	Kandahar	<NA>
3	AFG	SAS	Afghanistan	LIC	Mazar-E-...	<NA>
4	AFG	SAS	Afghanistan	LIC	Kabul	<NA>
5	AFG	SAS	Afghanistan	LIC	HiratÂ	<NA>
6	AGO	SSF	Angola	LMC	Luanda	<NA>
7	ALB	ECS	Albania	UMC	Korca	<NA>
8	ALB	ECS	Albania	UMC	Vlora	<NA>
9	ARE	MEA	United Arab Emira...	HIC	Abu Dhabi	<NA>
10	ARE	MEA	United Arab Emira...	HIC	Dubai	<NA>

i 357 more rows

i abbreviated name: ¹additional_data_annual_budget_for_waste_management_year

i 107 more variables: additional_data_annual_solid_waste_budget_year <chr>,

additional_data_annual_swm_budget_2017_year <dbl>,

additional_data_annual_swm_budget_year <dbl>.

Writing data as CSV files

- transform data
- export processed analysis-ready data

```
1 # data transformation
2 waste_sml <- waste |>
3   select(country_name, city_name, iso3c, income_id,
4           total_msw_total_msw_generated_tons_year,
5           population_population_number_of_people) |>
6   rename(country = country_name,
7           city = city_name,
8           generation_tons_year = total_msw_total_msw_generated_tons_year,
9           population = population_population_number_of_people)
10
11 # export processed analysis-ready data
12 write_csv(waste_sml, "data/processed/waste-city-level-sml.csv")
```

Reading data from XLSX files

- import unprocessed raw data

```
1 sludge <- read_excel("data/raw/tbl-01-faecal-sludge-analysis.xlsx",
2                       sheet = 1)
3 sludge
```

A tibble: 20 × 6

	id	date_sample	system	location	users	ts
	<dbl>	<dtm>	<chr>	<chr>	<dbl>	<dbl>
1	1	2023-11-01 00:00:00	pit latrine	household	5	136.
2	2	2023-11-01 00:00:00	pit latrine	household	7	102.
3	3	2023-11-01 00:00:00	pit latrine	household	NA	57.0
4	4	2023-11-01 00:00:00	pit latrine	household	6	27.0
5	5	2023-11-01 00:00:00	pit latrine	household	12	97.3
6	6	2023-11-02 00:00:00	septic tank	household	7	78.2
7	7	2023-11-02 00:00:00	septic tank	household	14	15.2
8	8	2023-11-02 00:00:00	septic tank	household	4	29.4
9	9	2023-11-02 00:00:00	septic tank	household	10	64.2
10	10	2023-11-02 00:00:00	septic tank	household	12	8.01
11	11	2023-11-03 00:00:00	pit latrine	public toilet	50	11.2
12	12	2023-11-03 00:00:00	pit latrine	public toilet	32	84.0
13	13	2023-11-03 00:00:00	pit latrine	public toilet	41	55.9
14	14	2023-11-03 00:00:00	pit latrine	public toilet	160	15.3
15	15	2023-11-03 00:00:00	pit latrine	public toilet	20	22.6

Writing data as CSV files

- transform data
- export data underlying a publication

```

1 # data transformation
2 tbl_sludge_summary <- sludge |>
3   filter(!is.na(users)) |>
4   group_by(system, location) |>
5   summarise(
6     count = n(),
7     mean_ts = mean(ts),
8     sd_ts = sd(ts),
9     median_ts = median(ts)
10  )
11
12 # export data underlying a publication
13 write_csv(tbl_sludge_summary, "data/final/tbl-01-faecal-sludge-summary.csv")

```

system	location	count	mean_ts	sd_ts	median_ts
pit latrine	household	4	90.7	45.9	99.9
pit latrine	public toilet	5	37.8	31.3	22.6
septic tank	household	5	39.0	30.8	29.4
septic tank	public toilet	5	20.4	14.3	15.6

(Research) Data Management

Examples of terms used when managing data.

term	folder	explanation	file format
unprocessed raw data	raw	data that is not processed and remains in its original form and file	often XLSX, also CSV and others
processed analysis-ready data	processed	data that is processed to prepare for an analysis and is exported in its new form as a new file	CSV
final data underlying a publication	final	data that is the result of an analysis (e.g. descriptive statistics or data visualization) and shown in a report, but then also exported in its new form as a new file	CSV

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Data Organization in Spreadsheets

Data Organization in Spreadsheets

The screenshot shows a web browser displaying the full article page for "Data Organization in Spreadsheets" on the Taylor & Francis Online platform. The browser's address bar shows the DOI: <https://doi.org/10.1080/00031305.2017.1375989>. The page header includes the Taylor & Francis Online logo, access provided by ETH-Bibliothek, and navigation links for Log in, Register, and Cart. The breadcrumb trail indicates the article is part of "The American Statistician", Volume 72, Issue 1, a Special Issue on Data Science. The article title "Data Organization in Spreadsheets" is prominently displayed, along with the authors Karl W. Broman & Kara H. Woo. The article is marked as "Open access". On the left, a sidebar shows the article's metrics: 318,984 Views, 56 CrossRef citations to date, and 2,110 Altmetric. Below the article title, there are links to "Full Article", "Figures & data", "References", "Citations", "Metrics", "Licensing", "Reprints & Permissions", and a "View PDF" button. The "ABSTRACT" section begins with the text: "Spreadsheets are widely used software tools for data entry, storage, analysis, and visualization. Focusing on the data entry and storage aspects, this article offers practical recommendations for organizing spreadsheet data to reduce". On the right, a "Related research" section includes buttons for "Recommended articles", "People also read", and "Cited by 56". A link "Using spreadsheets to teach" is visible at the bottom of the related research section. The bottom of the page features a footer with a GitHub link and the "open wash data." logo.

Full article: Data Organization x +

tandfonline.com/doi/full/10.1080/00031305.2017.1375989?src=

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318,984 Views

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Article

Data Organization in Spreadsheets

Karl W. Broman & Kara H. Woo

Pages 2-10 | Received 01 Jun 2017, Published online: 24 Apr 2018

Cite this article | <https://doi.org/10.1080/00031305.2017.1375989> | Check for updates

Open access

Full Article | Figures & data | References | Citations | Metrics | Licensing | Reprints & Permissions | View PDF

ABSTRACT

Spreadsheets are widely used software tools for data entry, storage, analysis, and visualization. Focusing on the data entry and storage aspects, this article offers practical recommendations for organizing spreadsheet data to reduce

Related research

Recommended articles | People also read | Cited by 56

Using spreadsheets to teach

Data Organization in Spreadsheets

Read the paper (it's part of your homework), but you can also:

- Go through the annotated slides:
https://kbroman.org/Talk_DataOrg/dataorg_notes.pdf
- Watch Karl Broman give the talk (02:36 to 45:00):
<https://youtu.be/t74E0a90gkA?t=156>
- Read the content on a website: <https://kbroman.org/dataorg/>

Data Organization in Spreadsheets

But, especially apply it to your data

Data Organization in Spreadsheets

Why? Because following a set of rules for organizing data everyone's live a little better.

The screenshot shows the 'Most read articles' section of The American Statistician website. The page lists three articles with their respective view counts, citations, and Altmetric scores. The third article, 'Data Organization in Spreadsheets', is highlighted with a pink dashed box.

Article Title	Views	CrossRef citations	Altmetric
The ASA Statement on p -Values: Context, Process, and Purpose	679927	3566	2,377
Moving to a World Beyond " $p < 0.05$ "	349365	1546	1,400
Data Organization in Spreadsheets	318984	56	2,110

- 3rd most viewed paper on The American Statistician
- 310'000+ views
- widely accepted as minimal standards

Data Organization in Spreadsheets

License? CC0 (!)

☰ README.md

Data organization in spreadsheets

Slides for a talk for the [OSGA Webinar Series](#), on 24 Sept 2021, based on [my paper of the same title with Kara Woo](#). Also see the [related website](#).

PDF of slides: https://kbroman.org/Talk_DataOrg/dataorg.pdf

PDF of slides with notes: https://kbroman.org/Talk_DataOrg/dataorg_notes.pdf

Video of presentation: <https://youtu.be/t74E0a90gkA>

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Homework assignments module 4

Assignment 1: Identify and assess data

! This is the most important assignment!

To support you efficiently through to the end of the course, we need to identify your capstone project data now.

This gives us time to discuss data suitability with each of you individually.

Your tasks:

1. Identify 1-2 potential datasets
2. Assess privacy and sensitivity
3. Document required modifications
4. Open GitHub issue to discuss with instructors

Remember:

- All capstone projects will be published openly
- Personal identifiers must be removed
- When in doubt, exclude or aggregate

Assignment 2: Study groups (Mentorship programme)

We've identified some participants with more experience who can help support others in the course.

Join a study group on GitHub Discussions!

- Groups will have 5-6 members
- Meet with your group to discuss
- Support each other through the modules

1. Navigate to [GitHub Discussions](#)
2. Find the **Group** category
3. **Join one of the discussions** to join a group
4. Be mindful to distribute evenly - aim for 5-6 members per group

⚠ Important

If your username appears in the assignment instructions, please join the study group with your username in the title - you've been identified as someone with experience who can help others!

Module 4 documentation

ds4owd-002.github.io/website/content/modules/md-04.html

Homework due date

- Homework assignment due: Wednesday, October 8th 2025
- Quiz due: Wednesday, October 15th 2025

Course Calendar

date	week	topic	module
11 September 2025	1	Welcome & get ready for the course	module 1
18 September 2025	2	Data science lifecycle & Exploratory data analysis using visualization	module 2
25 September 2025	3	Data transformation with dplyr	module 3
02 October 2025	4	Data import & Data organization in spreadsheets	module 4
09 October 2025	5	No class	NA
16 October 2025	6	No class	NA
23 October 2025	7	Conditions & Dates & Tables	module 5
30 October 2025	8	Data types & Vectors & For Loops	module 6
06 November 2025	9	Pivoting & joining data	module 7
13 November 2025	10	Creating and publishing scholarly articles with Quarto and GitHub pages	module 8
20 November 2025	11	Bonus module: Use of AI for coding support	module 9
27 November 2025	12	Work on Capstone project	NA
04 December 2025	13	Work on Capstone project	NA
11 December 2025	14	Final submission date of Capstone project	NA
18 December 2025	15	Graduation of openwashdata academy	module 10

Wrap-up

Thanks!

Slides created via revealjs and Quarto:

<https://quarto.org/docs/presentations/revealjs/>

Access slides as [PDF on GitHub](#)